

Weekly Progress Report

Project: Prediction of Agriculture Crop Production In India

Organization: Upskill Campus

Intern: MUKUL VERMA **Week:** 2 **Date:** 08/07/2026

1. TASKS COMPLETED & MILESTONES

During this week, I worked on three modules of the Smart Farming Assistant project using Google Colab and Python.

Model 1: Yield Prediction

- Imported and cleaned *datafile (1).csv* using Pandas.
- Checked missing values and prepared the dataset for analysis.
- Built a script that accepts State and Crop as user input.
- Calculated average Yield and Cost from the filtered records.
- Tested the script with sample inputs and exported the results to a CSV file.

Model 2: Production Forecast

- Loaded *datafile (2).csv* and *produce.csv*.
- Merged the datasets using `pd.concat()`.
- Displayed crop-wise and year-wise production details.
- Generated a production trend graph and predicted production values for 2026.
- Evaluated predictions using R^2 Score and Mean Absolute Error (MAE).

Model 3: Variety and Season Recommendation

- Loaded *datafile (3).csv*.
- Solved parsing issues using `engine='python'` and `on_bad_lines='skip'`.
- Removed the unnecessary *Unnamed:4* column.
- Developed a recommendation script that suggests suitable crop varieties and season duration based on State and Crop.
- Saved recommendation results as a CSV report.

Milestone Achieved: All three modules were completed successfully and produced the expected outputs.

2. SIGNIFICANT CONTRIBUTIONS

- Organized five agricultural datasets and mapped each dataset to the appropriate module.
- Implemented data cleaning and preprocessing before generating results.
- Developed a single menu-driven interface to execute all three modules from one Python program.
- Generated prediction outputs, recommendation reports, evaluation metrics, and CSV files.

3. CHALLENGES AND HURDLES

Challenge 1: Errors while reading *datafile (3).csv* because of bad rows and an extra column.

Solution: Used `engine='python'`, `on_bad_lines='skip'`, and removed the *Unnamed:4* column.

Challenge 2: Difficulty in deciding which dataset should be used for each module.

Solution: Prepared a mapping of datasets and verified each module before implementation.

Challenge 3: Ensuring accurate user input filtering and correct CSV generation.

Solution: Tested the scripts with different State and Crop combinations and validated the outputs.

4. LESSONS LEARNED

- Learned to preprocess and clean multiple CSV datasets using Pandas.
- Improved understanding of filtering, aggregation, and CSV export.
- Learned how to evaluate prediction models using R^2 Score and MAE.
- Gained experience in generating graphs for production analysis.
- Improved debugging and error-handling skills.
- Learned to organize multiple modules into a single Python application.

PROGRESS REPORT OUTPUT:

Model 1

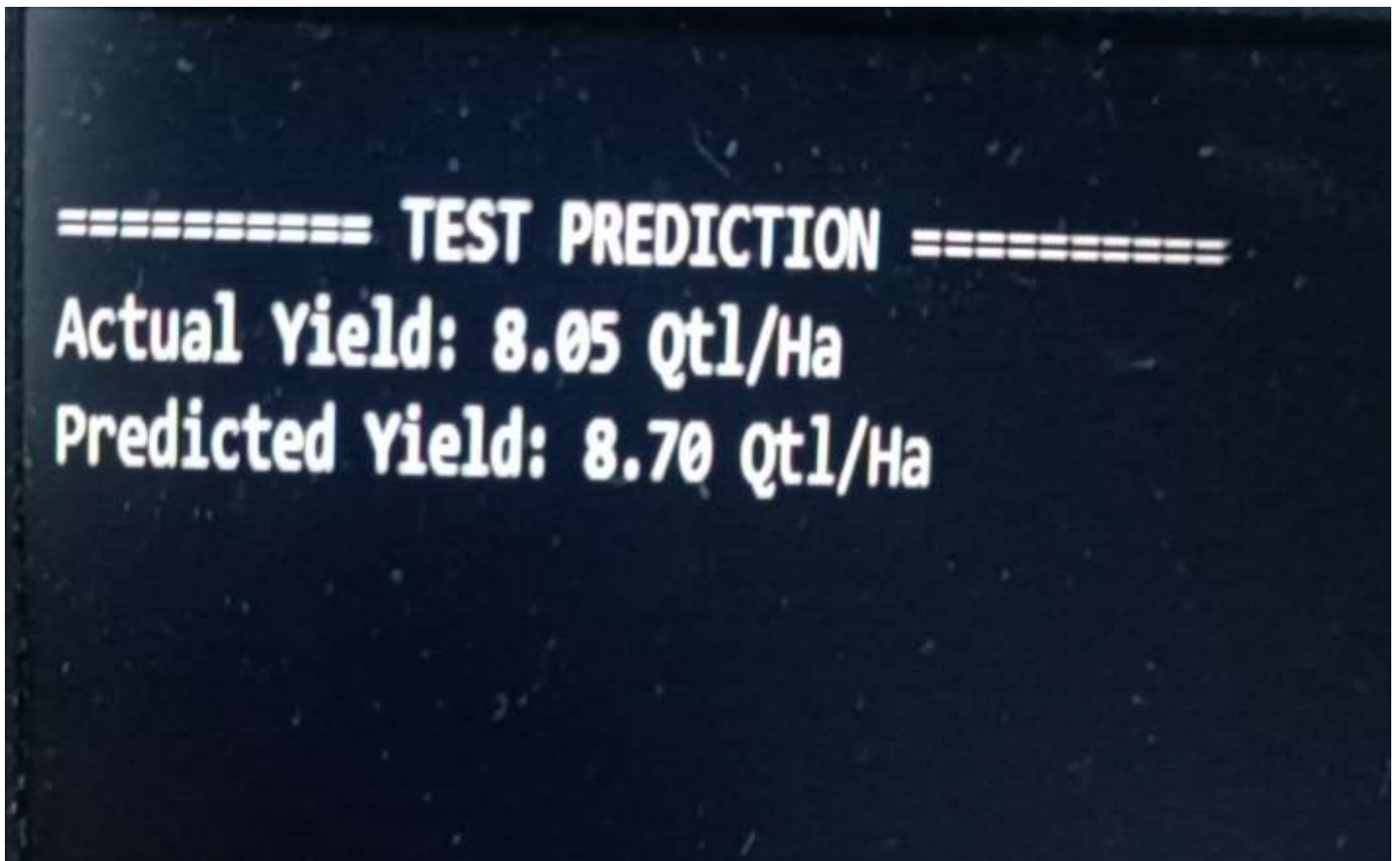


Figure 1: Test Prediction Output (Actual Yield vs Predicted Yield)

Model 2

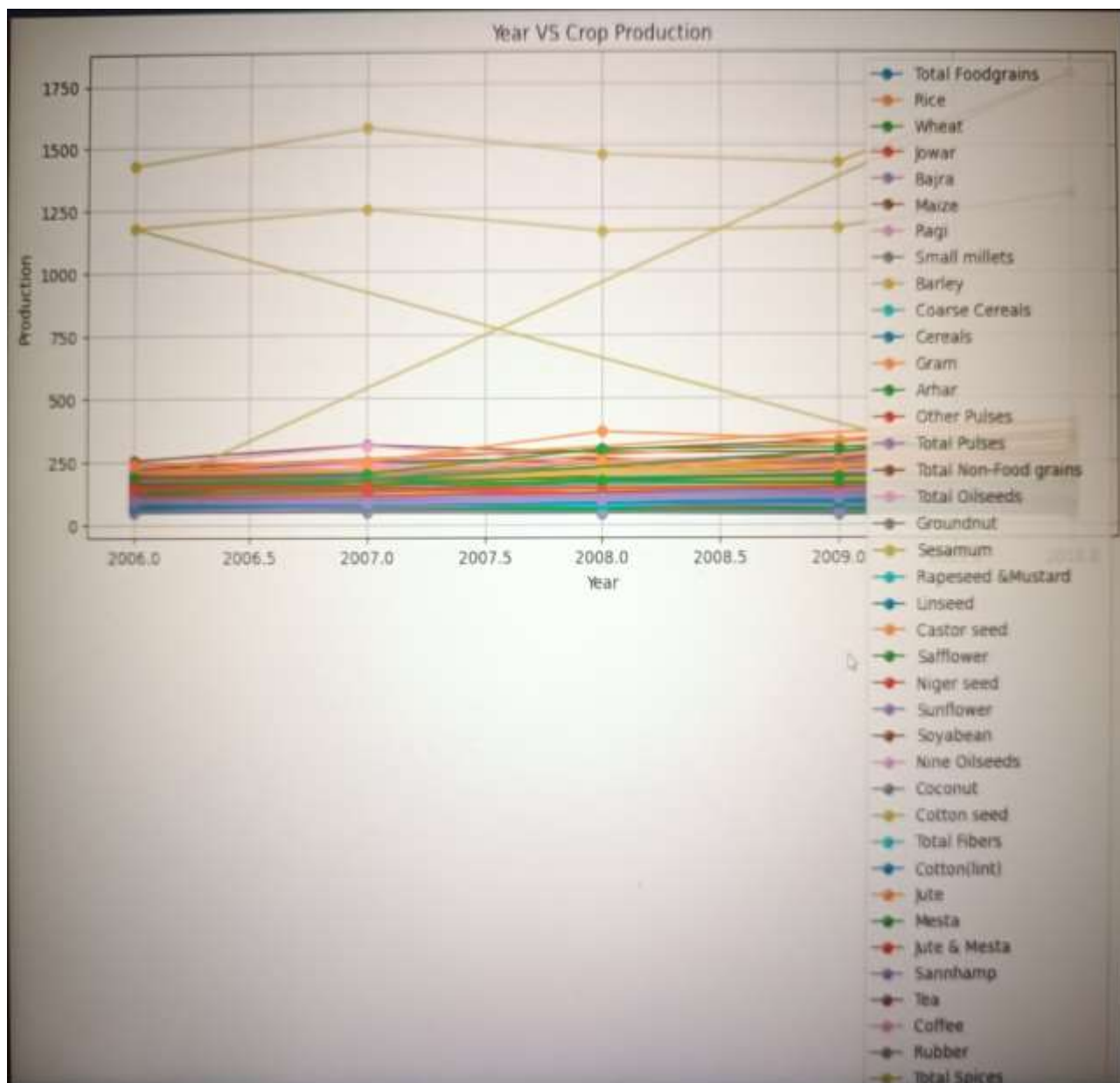


Figure 2: Year vs Crop Production Graph

```

===== 2026 PREDICTION =====
Total Foodgrains 2026: 174.93
Rice 2026: 157.01
Wheat 2026: 172.49
Jowar 2026: 85.93
Bajra 2026: 138.88
Maize 2026: 326.78
Ragi 2026: 157.79
Small millets 2026: 167.62
Barley 2026: 170.05
Coarse Cereals 2026: 196.81
Cereals 2026: 174.76
Gram 2026: 267.49
Arhar 2026: 131.28
Other Pulses 2026: 180.0
Total Pulses 2026: 205.91
Total Non-Food grains 2026: 147.13
Total Oilseeds 2026: 159.82
Groundnut 2026: 219.49
Sesamum 2026: 168.89
Rapeseed & Mustard 2026: 171.76
Linseed 2026: 70.97
Castor seed 2026: 343.12
Safflower 2026: -4.14
Niger seed 2026: 51.21
Sunflower 2026: -145.67
Soyabean 2026: 413.79
Nine Oilseeds 2026: 216.79
Coconut 2026: 11.6
Cotton seed 2026: 319.37
Total Fibers 2026: 259.09
Cotton(lint) 2026: 353.79
Jute 2026: 191.34
Mesta 2026: -33.23
Jute & Mesta 2026: 165.38
Sannhamp 2026: 157.81
Tea 2026: 121.63
Coffee 2026: 196.26
Rubber 2026: 132.09
Total Spices 2026: 1441.09
Black pepper 2026: 33.57
Dry chillies 2026: 122.79
Dry ginger 2026: 502.85
Turmeric 2026: 229.71
Arecanut 2026: 162.71
Cardamom 2026: 187.34
Coriander 2026: 366.67
Garlic 2026: 298.27
Total Fruits & Vegetables 2026: 457.27
Potato 2026: 465.23

```

Figure 3: Production Forecast for the Year 2026


```

Total Foodgrains -> R2: 0.019 | MAE: 16.9
Rice -> R2: 0.0 | MAE: 28.92
Wheat -> R2: 0.108 | MAE: 9.6
Jowar -> R2: 0.066 | MAE: 7.33
Bajra -> R2: 0.0 | MAE: 30.12
Maize -> R2: 0.059 | MAE: 38.69
Ragi -> R2: 0.065 | MAE: 19.11
Small millets -> R2: 0.034 | MAE: 31.64
Barley -> R2: 0.05 | MAE: 21.66
Coarse Cereals -> R2: 0.031 | MAE: 23.27
Cereals -> R2: 0.013 | MAE: 18.36
Gram -> R2: 0.266 | MAE: 13.86
Arhar -> R2: 0.031 | MAE: 9.12
Other Pulses -> R2: 0.044 | MAE: 15.24
Total Pulses -> R2: 0.179 | MAE: 11.37
Total Non-Food grains -> R2: 0.004 | MAE: 13.14
Total Oilseeds -> R2: 0.021 | MAE: 11.92
Groundnut -> R2: 0.037 | MAE: 24.27
Sesamum -> R2: 0.037 | MAE: 22.16
Rapeseed & Mustard -> R2: 0.044 | MAE: 16.06
Linseed -> R2: 0.0 | MAE: 32.49
Castor seed -> R2: 0.22 | MAE: 22.88
Safflower -> R2: 0.031 | MAE: 31.87
Niger seed -> R2: 0.012 | MAE: 13.32
Sunflower -> R2: 0.399 | MAE: 20.73
Soyabean -> R2: 0.037 | MAE: 67.31
Nine Oilseeds -> R2: 0.066 | MAE: 17.82
Coconut -> R2: 0.125 | MAE: 16.49
Cotton seed -> R2: 0.095 | MAE: 30.71
Total Fibers -> R2: 0.052 | MAE: 28.16
Cotton(lint) -> R2: 0.068 | MAE: 41.56
Jute -> R2: 0.0 | MAE: 49.55
Mesta -> R2: 0.097 | MAE: 24.62
Jute & Mesta -> R2: 0.0 | MAE: 48.87
Sannhamp -> R2: 0.003 | MAE: 54.98
Tea -> R2: 0.0 | MAE: 17.1
Coffee -> R2: 0.147 | MAE: 11.37
Rubber -> R2: 0.006 | MAE: 34.39
Total Spices -> R2: 0.004 | MAE: 554.78
Black pepper -> R2: 0.116 | MAE: 11.95
Dry chillies -> R2: 0.002 | MAE: 41.02
Dry ginger -> R2: 0.158 | MAE: 44.07
Turmeric -> R2: 0.097 | MAE: 17.39
Areca nut -> R2: 0.0 | MAE: 31.32
Cardamom -> R2: 0.008 | MAE: 36.97
Coriander -> R2: 0.149 | MAE: 35.81
Garlic -> R2: 0.015 | MAE: 52.9
Total Fruits & Vegetables -> R2: 0.248 | MAE: 32.2
Potato -> R2: 0.312 | MAE: 29.93

```

Figure 4: Model Performance Evaluation (R^2 Score and MAE)

Model 3

```
===== CROP VARIETY RECOMMENDER =====
Available Crops: ['Paddy' 'Wheat' 'Barley' 'Maize' 'Pearl Millet' 'Finger Millet'
'Indian Mustard' 'Yellow Sarson' 'Linseed' 'Groundnut' 'Sesame'
'Chickpea' 'Bengal Gram' 'Lentil' 'Fieldpea' 'French Bean' 'Mungbean'
'Urdbean' 'Cluster Bean' 'Horse Gram' 'Sugarcane' 'Napier Bajra Hybrid'
'Oat' 'Tall Fescue Grass' 'Cowpea (Fodder)' 'Jute' 'Mesta' 'Cotton'
'Desi Cotton']

Enter State name: Rajasthan
Enter Crop name: Pearl Millet

Found 5 varieties for 'Pearl Millet' in Rajasthan:

=====
Crop          Variety Season_Duration      Recommended_Zone
Pearl Millet  Nandi-65 (MH-1549)          80 Rajasthan, Gujarat, Haryana, MP, UP, Punjab and Delhi in Kharif season.
Pearl Millet  Nandi-61 (MH-1548)          81 Rajasthan, Gujarat, Haryana, Punjab and Delhi in Kharif season.
Pearl Millet  86M64 (MSH 283) (Hybrid)     80-85 Rajasthan, Gujarat, Maharashtra and Tamil Nadu in summer season.
Pearl Millet  RHB 177 (MH 1486)           NaN Rajasthan, Haryana and Gujarat under rainfed low fertility condition.
Pearl Millet  HMB 226 (MH 1479)           NaN Rajasthan, Gujarat, Haryana, Uttar Pradesh, Punjab and Madhya Pradesh.

File created: Rajasthan_Pearl Millet_Recommendation.csv
```

Figure 5: Crop Variety Recommendation Output